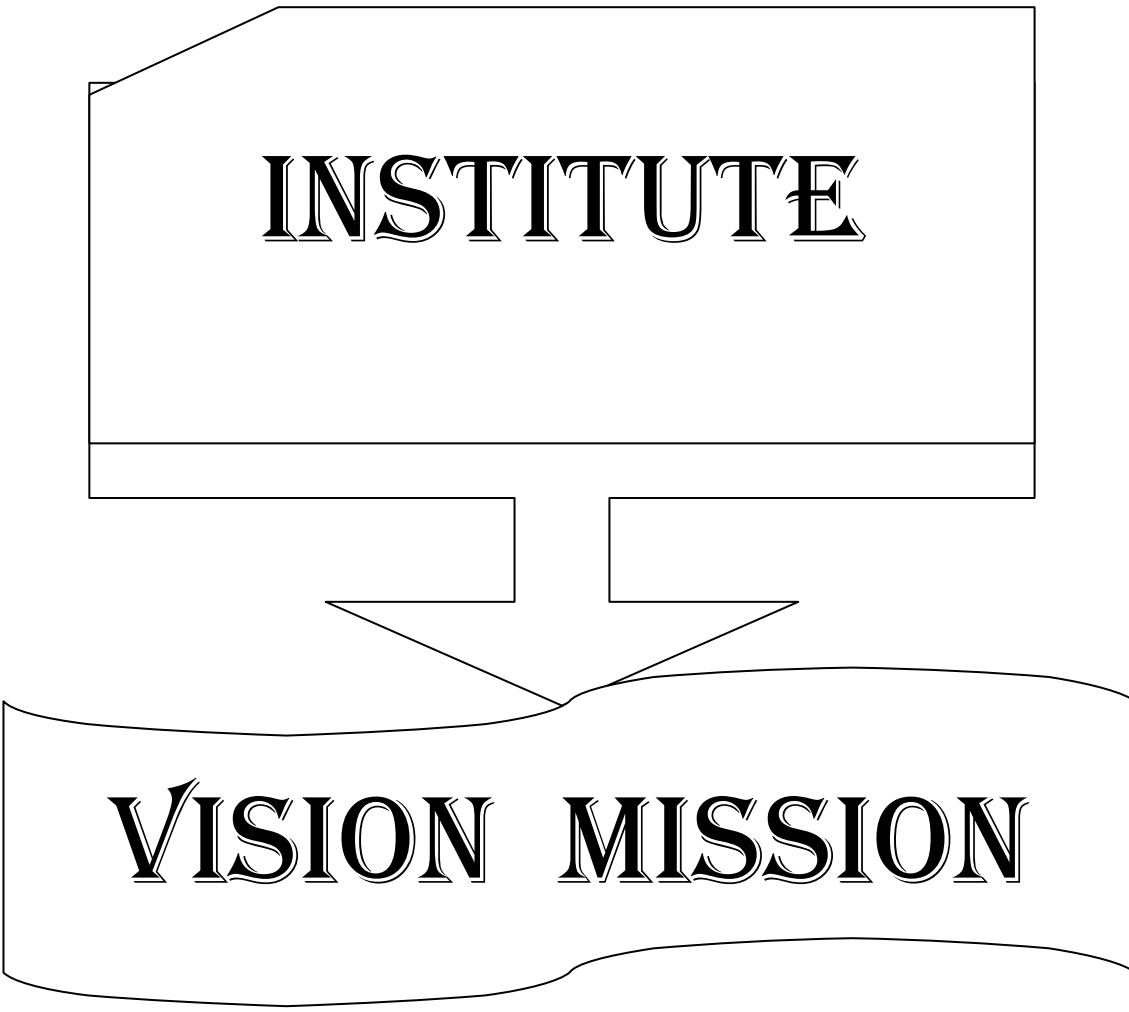


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INSTITUTE

VISION MISSION

INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning..

DEPARTMENT

VISION

MISSION

DEPARTMENT

VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

**PROGRAM
EDUCATIONAL
OBJECTIVES,
PROGRAM OUTCOMES
& PROGRAM
SPECIFIC
OUTCOMES**

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. **[K3]**
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. **[K4]**
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. **[K5]**
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **[K5]**
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. **[K3]**
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. **[K3]**
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. **[K3]**
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. **[K3]**
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. **[K6]**
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[K2]**
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. **[K6]**
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. **[K1]**

ACADEMIC CALENDAR

A.Y. 2019-20

B.TECH III SEM			
DESCRIPTION	FROM	TO	WEEKS
COMMENCEMENT OF CLASS WORK	10.06.2019		
I UNIT OF INSTRUCTIONS	10.06.2019	03.08.2019	8 WEEKS
I MID EXAMINATION	05.08.2019	10.08.2019	1 WEEK
II UNIT OF INSTRUCTIONS	12.08.2019	05.10.2019	8 WEEKS
II MID EXAMINATION	07.10.2019	12.10.2019	1 WEEK
COMPREHENSIVE TEST	14.10.2019	19.10.2019	1 WEEK
PREPARATION AND PRACTICALS	21.10.2019	25.10.2019	5 DAYS
END EXAMINATIONS	28.10.2019	09.11.2019	2 WEEKS
B.TECH IV SEM			
COMMENCEMENT OF CLASS WORK	11.11.2019		
I UNIT OF INSTRUCTIONS	11.11.2019	04.01.2020	8 WEEKS
I MID EXAMINATION	06.01.2020	11.01.2020	1 WEEK
II UNIT OF INSTRUCTIONS	13.01.2020	07.03.2020	8 WEEKS
II MID EXAMINATION	09.03.2020	14.03.2020	1 WEEK
COMPREHENSIVE TEST	16.03.2020	21.03.2020	1 WEEK
PREPARATION AND PRACTICALS	23.03.2020	27.03.2020	5 DAYS
END EXAMINATIONS	30.03.2020	11.04.2020	2 WEEKS

CO-CURRICULAR AND EXTRA CURRICULAR ACTIVITIES

05/06/2019 – World Environment Day	15/09/2019 – Engineers Day		21/03/2019 – International Forest Day
21/08/2019 to 24/08/2019– 1st year B.Tech. Introduction program	16/09/2019 – World Ozone Day	22/12/2019 – National Mathematics Day	22/03/2019 –World Water Day
15/08/2018 – Independence Day	September – Intramurals	26/01/2019 – Republic Day	23/03/2019 – World Meteorological Day
05/09/2019 – Teachers Day	11/11/2019 – National Education Day	08/03/2019 – International Women’s Day	24/03/2019 – Earth Hour
	03/12/2019 – Antipollution Day		In the month of March – Association Days



SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



CLASS CONSOLIDATED TIME TABLE

Class: 3rd Semester

Section - A

Class Coordinator :Mrs.A.Sirisha

w.e.f. 10-06-2019

Room No:PG-202

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	DM	DSA	PCS-I		LUNCH BREAK	OOPS	DE	P&S
Tue	DE	DS LAB				DM	OOPS	PCS-I
Wed	DSA	P&S	TS-I	DE		DE LAB		
Thu	P&S	DE	DM	DSA		OOPS LAB		
Fri	DSA	TS-I LAB				P&S	OOPS	DM
Sat	DE	OOPS	DSA	P&S		OOPS	DM	LIB/SPO

Section - B

Class Coordinator: Mrs.B.SriRamya

Room No: PG-203

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	DSA	DE LAB			LUNCH BREAK	P&S	OOPS	DM
Tue	OOPS	DSA	DM	DE		PCS-I		LIB/SPO
Wed	DM	OOPS	DE	P&S		DSA	DE	PCS-I
Thu	P&S	OOPS LAB				P&S	DM	DE
Fri	TS-I	OOPS	DSA	P&S		DS LAB		
Sat	DE	TS-I LAB				DM	DSA	OOPS

Section - C

Class Coordinator: Mrs.N.Hiranmayee

Room No: PG-204

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	DSA	OOPS LAB			LUNCH BREAK	DM	P&S	PCS-I
Tue	DM	DE	DSA	OOPS		DS LAB		
Wed	DE	TS-I LAB				OOPS	DM	P&S
Thu	P&S	DM	P&S	TS-I		DE	DSA	OOPS
Fri	DM	DE	P&S	OOPS		DE	DSA	LIB/SPO
Sat	OOPS	DSA	PCS-I			DE LAB		

Section - D

Class Coordinator: Mr.O.Sri Nagesh

Room No: PG-205

Periods	1	2	3	4	1:00PM 2:00PM LUNCH BREAK	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	DE	OOPS	DSA	DM		DS LAB		
Tue	DM	P&S	DE	OOPS		P&S	DSA	DE
Wed	DSA	OOPS LAB				P&S	DM	OOPS
Thu	OOPS	TS-I LAB				DSA	DE	DM
Fri	P&S	DE LAB				PCS-I		LIB/SPO
Sat	DSA	DM	DE	OOPS		PCS-I	P&S	TS-I

Staff Details:

S. No.	Course Code	Course Name	Section			
			A	B	C	D
1.	V18MAT04	P&S	Mrs.B.Adilakshmi	Mr.S.K.Dhanaprasad	Mrs.G.S.Prasanthi	Mrs.B.Adilakshmi
2.	V18ECT06	DE	Mr.P.V.V.Satyanarayana	Mr.J.Rajendra	Mr.J.Rajendra	Mr.P.V.V.Satyanarayana
3.	V18CST02	DSA	Mr.Ch.RajaRamesh	Mrs.B.Sri Ramya	Mrs.A.Sirisha	Mrs.B.Sri Ramya
4.	V18CST03	DM	Dr.V.S.Naresh	Mrs.N. Hiranmayee	Mrs.N. Hiranmayee	Dr.V.S.Naresh
5.	V18CST04	OOPS	Mr.O.Sri Nagesh	Dr.K.ShirinBhanu	Dr.K.ShirinBhanu	Mr.O.Sri Nagesh
6.	V18ECL04	DE Lab	Mr.P.V.V.Satyanarayana, Mr. R.L.R. Lokeshbabu, Mr. K. RAJESH, Mr. T.Vijaya Sai	Mr.P.V.V.Satyanarayana Mr. R.L.R. Lokeshbabu, Mr.J.Rajendra, Sri N.V. NarayanaRao	Mr.P.V.V.Satyanarayana, Mr. R.L.R. Lokeshbabu, Mr.J.Rajendra, Sri N.V. NarayanaRao	Mr.P.V.V.Satyanarayana, Mr. R.L.R. Lokeshbabu, Mr. T.Vijaya Sai, Mr. A.R.S. Balaji
7.	V18CSL02	DSA Lab	Mr.Ch.RajaRamesh, Mr.K.Lakshmi Narayana Mrs.K.Manasa	Mrs.B.Sri Ramya, Mrs.A.L.Lavanya	Mrs.A.Sirisha, Ms.I.Lavanya	Mrs.B.Sri Ramya, Ms.P.Sukanya
8.	V18CSL03	OOPS Lab	Mr.O.Sri Nagesh Mr.N.Praveen Kumar	Dr.K.ShirinBhanu, Sd.Sabir Hussain	Dr.K.ShirinBhanu, Mrs.K.Rajani Kumari	Mr.O.Sri Nagesh Mr.K.Lakshmi Narayana
9.	V18CST60	TS-I	Mrs.A.Sirisha	Mr.G.Nataraj	Mr.G.Nataraj	Mrs.A.Sirisha
10.	V18ENT03	PCS-I	Mr.D.Naveen Kumar Mr.B.Anjaneyulu	Mr.D.Naveen Kumar Mr.B.Anjaneyulu	Mr.D.Naveen Kumar Mr.B.Anjaneyulu	Mr.D.Naveen Kumar Mr.B.Anjaneyulu



Head of the Department

COURSE STRUCTURE

III-Semester

B.TECH Course Structure

S.No.	III - Semester						
	Course Code		Course	L	T	P	C
1	V18MAT04	BSC	Probability & Statistics	3	1	0	4
2	V18ECT06	ESC	Digital Electronics	3	0	0	3
3	V18CST02	PCC	Data Structures and Algorithms	3	0	0	3
4	V18CST03	ESC	Discrete Mathematics	3	0	0	3
5	V18CST04	ESC	Object Oriented Programming for problem Solving	3	0	0	3
6	V18ECL04	ESC	Digital Electronics Lab	0	0	2	1
7	V18CSL02	PCC	Data Structures and Algorithms Lab	0	0	3	1.5
8	V18CSL03	ESC	Object Oriented Programming for problem Solving Lab	0	0	3	1.5
9	V18ENT03		Professional Communication Skills – I	3	0	0	MNC
10	V18CST60		Technical Skills-I	0	0	4	MNC
Total				18	1	12	20

Total Contact Hours: 31



LESSON PLANS

PROBABILITY AND STATISTICS

Academic Year: 2019-20

Semester: III

Name of the Course: PROBABILITY AND STATISTICS

Programme: B.Tech

Section: A,B,C& D

Course Code: V18MAT04

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, Student will be able to:

	Course Outcome	Knowledge level of the CO
CO1	Find measures of central tendency and dispersion for real data sets.	K3
CO2	Find parameters of given function	K3
CO3	Apply probability distribution to real time problems	K3
CO4	Plot a best fit curve to an experimental data and find the correlation and regression	K3
CO5	Create good estimators to various parameters	K6
CO6	Apply the principles of Statistical Inference to practical problems	K3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
		Unit – 1: Basic Statistics	K3	10		
1	CO1	Define measures of central tendency or average , mean	K1	1	Lecture method	Black Board
2	CO1	Find mean of the given data and discuss its merits and demerits	K3	1	Lecture method	Black Board
3	CO1	Explain median, merits and demerits of median	K2	1	Lecture method	Black Board
4	CO1	Find median of the ungrouped and grouped data	K3	1	Lecture method	Black Board
5	CO1	Explain mode, merits and demerits of median	K2	1	Lecture method	Black Board
6	CO1	Find mode of the ungrouped and grouped data	K3	1	Lecture method	Black Board
7	CO1	Define measures of dispersion: standard deviation and variance	K1	1	Lecture method	Black Board
8	CO1	Find variance and standard deviation	K3	1	Lecture method	Black Board
9	CO1	Define kurtosis and skewness	K1	1	Lecture method	Black Board

10	CO1	Find kurtosis and Skewness	K3	1	Lecture method	Black Board
		UNIT-II: Basic Probability	K3	10		
11	CO2	Define random variables: discrete and continuous with examples	K1	1	Lecture method	Black Board
12	CO2	Explain discrete probability distribution, probability function, density function and their properties	K2	1	Lecture method	Black Board
13	CO2	Explain expectation , mean , variance, standard deviation of a probability distribution	K2	1	Lecture method	Black Board
14	CO2	Find the probabilities using discrete probability function	K3	1	Lecture method	Black Board
15	CO2	Calculate expectation , mean , variance and standard deviation of discrete distribution	K3	1	Lecture method	Black Board
16	CO2	Explain continuous probability distribution, probability function, density function and their properties	K2	1	Lecture method	Black Board
17	CO2	Find the probabilities using continuous probability function	K3	1	Lecture method	Black Board
18	CO2	Find the mean, median , mode , variance and standard deviation of continuous probability distribution	K3	1	Lecture method	Black Board
19	CO2	Explain moments and Chebyshev's Inequality	K2	1	Lecture method	Black Board
20	CO2	Find moments of given data	K3	1	Lecture method	Black Board
		UNIT III: Probability Distributions	K3	10		
21	CO3	Explain Binomial distribution and its properties	K2	1	Lecture method	Black Board
22	CO3	Find the probability using Binomial distribution	K3	1	Lecture method	Black Board
23	CO3	Explain Poisson distribution and its properties	K2	1	Lecture method	Black Board
24	CO3	Find the probability using Poisson distribution	K3	1	Lecture method	Black Board
25	CO3	Explain Normal distribution and its properties	K2	1	Lecture method	Black Board
26	CO3	Find the probabilities by using Normal distribution	K3	1	Lecture method	Black Board
27	CO3	Find the probabilities by using Normal distribution	K3	1	Lecture method	Black Board
28	CO3	Explain Exponential distribution and weibull distribution	K2	1	Lecture method	Black Board

29	CO3	Explain Gamma distribution	K2	1	Lecture method	Black Board
30	CO3	Find the probabilities using Exponential, Weibull and Gamma distributions	K3	1	Lecture method	Black Board
		UNIT IV: Bivariate Distribution	K3	10		
31	CO4	Explain Least-Squares method to fit a curve to the given data	K2	1	Lecture method	Black Board
32	CO4	Apply the Method of Least-Squares to fit a straight line for the given data	K3	1	Lecture method	Black Board
33	CO4	Apply the Method of Least-Squares to fit a Second degree curve for the given data	K3	1	Lecture method	Black Board
34	CO4	Apply the Method of Least-Squares to fit a Exponential and power curve for the given data	K3	1	Lecture method	Black Board
35	CO4	Define correlation and types of correlation, correlation coefficient	K1	1	Lecture method	Black Board
36	CO4	Calculate Karl Pearson's coefficient of correlation for given data	K3	1	Lecture method	Black Board
37	CO4	Discuss Spearman's rank correlation coefficient	K2	1	Lecture method	Black Board
38	CO4	Find the rank correlation coefficient	K3	1	Lecture method	Black Board
39	CO4	Discuss regression	K2	1	Lecture method	Black Board
40	CO4	Calculate the regression equation for given data	K3	1	Lecture method	Black Board
		UNIT-V: Sampling Distribution and Estimation	K3	10		
41	CO5	Define population and samples, sampling theory	K1	1	Lecture method	Black Board
42	CO5	Define sampling distribution of means (σ known)	K1	1	Lecture method	Black Board
43	CO5	Calculate mean, variance and standard deviations of sampling distributions of means	K3	1	Lecture method	Black Board
44	CO5	Define sampling distribution of mean (σ unknown)	K1	1	Lecture method	Black Board
45	CO5	Explain t-distribution, F-	K2	1	Lecture	Black

		distribution and chi-square distribution			method	Board
46	CO5	Explain Estimation, criteria of a good estimator , point and interval estimation	K2	1	Lecture method	Black Board
47	CO5	Define the maximum error and confidence interval for the mean of a populations	K1	1	Lecture method	Black Board
48	CO5	Find the maximum error	K3	1	Lecture method	Black Board
49	CO5	Compute the confidence interval for the mean of a population using various distributions	K3	1	Lecture method	Black Board
50	CO5	Compute the confidence interval for the mean of a population using various distributions	K3	1	Lecture method	Black Board
		UNIT-VI Tests of Hypothesis	K3	13		
51	CO6	Define Null and alternative Hypothesis	K1	1	Lecture	Black Board
52	CO6	Explain Type-1,Type II errors and one tail and two tail test	K2	1	Lecture method	Black Board
53	CO6	Examine the hypothesis concerning mean and proportion using z-test	K3	1	Lecture method	Black Board
54	CO6	Examine the hypothesis concerning mean and proportion using z-test	K3	1	Lecture method	Black Board
55	CO6	Examine the hypothesis concerning two means and their differences	K3	1	Lecture method	Black Board
56	CO6	Examine the hypothesis concerning two proportions and their differences	K3	1	Lecture method	Black Board
57	CO6	Examine the hypothesis concerning small samples using t-test	K3	1	Lecture method	Black Board
58	CO6	Examine the hypothesis concerning small samples using t-test	K3	1	Lecture method	Black Board
59	CO6	Examine the hypothesis concerning small samples using t-test	K3	1	Lecture method	Black Board
60	CO6	Examine the hypothesis concerning variance using F-test	K3	1	Lecture method	Black Board
61	CO6	Examine the hypothesis concerning variance using F-test	K3	1	Lecture method	Black Board
62	CO6	Examine the hypothesis using chi-square test	K3	1	Lecture method	Black Board
63	CO6	Examine the hypothesis using chi-square test	K3	1	Lecture method	Black Board

Total No. of Classes: 63

Digital Electronics

Academic Year: 2019-20

Year/ Semester: III

Name of the Course: Digital Electronics

Programme: B.Tech

Section: A,B,C& D

Course Code: V18ECT06

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, Student will be able to:

CO	Course Outcome	Knowledge Level
CO1	Illustrate the conversion of a number from one number system to another.	K3
CO2	Classify Boolean theorems & simplify the Boolean functions using the Boolean properties.	K2
CO3	Use K-map as a tool to simplify and design logic circuits	K3
CO4	Construct different combinational Logic circuits like MUX, Decoders, Encoders etc.	K3
CO5	Demonstrate the basic flip-flops in terms of truth table & excitation table	K3
CO6	Apply the concepts of flip-flops in the designing of different sequential circuits like registers, counters, etc.	K3

Text Books/ Reference Books suggested:

TEXT BOOKS:

Text Books:

T1. Digital Design, 4/e, M.Morris Mano, Michael D Ciletti, PEARSON

T2. Fundamentals of Logic Design, 5/e, Roth ,Cengage

Reference books:

R1. Switching and Finite Automata Theory, 3/e, Kohavi, Jha, Cambridge.

R2. Digital Logic Design, Leach Malvino, Saha, TMH

R3. Modern Digital Electronics, R,P. Jain, TMH

UNIT1: Number systems& Binary codes:						
S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1		Dissemination of vision, mission, PEOs, POs, PSOs		1	Lecture	
2	CO 1	Define a Digital System & Number System	K1	1	Lecture	BB
3		Interpret Binary, Octal, Decimal, Hexadecimal Number Systems	K2	2	Lecture with Discussion	BB
4		Change Numbers From One Radix To Another Radix	K3	2	Lecture with Discussion	BB
5		Practice r's complement and (r-1)'s complement numbers	K3	2	Lecture with Discussion	BB
6		Illustrate Binary Codes	K3	2	Lecture with Discussion	BB + ICT
		TOTAL		10		

UNIT -II: Concept of Boolean algebra:						
S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Identify Basic Theorems	K1	1	Lecture	BB
2		Explain Properties of Boolean algebra	K2	1	Lecture With Discussion	BB
3		Explain Boolean Functions	K2	1	Lecture With Discussion	BB
4		Describe the Canonical and Standard Forms	K2	1	Lecture With Discussion	BB

5		Solve Minterms and Maxterms	K2	1	Lecture With Discussion	BB
6		Explain the Basic Logic Gates	K2	2	Lecture With Discussion	BB
7		Construct Universal Gates NAND and NOR, Ex-OR and Ex-NOR Gates.	K2	1	Lecture With Discussion	BB
8		Solve Problems	K2	1	Lecture With Discussion	BB
		TOTAL		9		

UNIT- III: Gate level Minimization:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Define Map Method,	K1	1	Lecture	BB
2		Differentiate 2,3,4-Variable K-Maps.	K2	2	Lecture With Discussion	BB
3		Estimate K-MAP for Products of Sum Simplification	K2	2	Lecture With Discussion	BB
4		Use K-MAP for Sum of Products Simplification	K3	2	Lecture With Discussion	BB
5		Find the SOP & POS forms for Don't-Care Conditions,	K3	2	Lecture With Discussion	BB
6		Develop NAND and NOR Implementation Exclusive OR function.	K3	1	Lecture With Discussion	BB
		TOTAL		10		

UNIT- IV: Combinational Logic						
S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Define & Find the combinational logic circuits	K2	1	Lecture With Discussion	BB + ICT
2		Dramatise the Analysis Procedure of Combinational logic circuit	K3	1	Lecture With Discussion	BB + ICT
3		Half adder ,Subtractor Full adder, Full Subtractor & RCA	K2	1	Lecture With Discussion	BB + ICT
4		Construct Binary Adder-Subtractor	K3	3	Lecture With Discussion	BB + ICT
5		Explain Decimal Adder	K3	1	Lecture With Discussion	BB + ICT
6		Construct Decoders-encoders	K3	2	Lecture With Discussion	BB + ICT
7		Construct Multiplexers & De-Multiplexers	K3	3	Lecture With Discussion	BB + ICT
		TOTAL		11		

UNIT V: Sequential Logic Circuits						
S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Define Latch & Flip -Flop	K1	1	Lecture With Discussion	BB + ICT
2		Explain-Latches and Basic Flip flops	K2	1	Lecture With Discussion	BB + ICT
3		Demonstrate RS, D, JK and T Flip-flop	K3	3	Lecture With Discussion	BB + ICT

4		Illustrate Clocked Sequential Circuits	K2	3	Lecture With Discussion & Seminar	BB + ICT
5		Change one Flip flop into another Flip-flop	K3	2	Lecture With Discussion	BB + ICT
		TOTAL		10		

UNIT -VI: Registers and Counters

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Explain Registers	K2	1	Lecture With Discussion	BB + ICT
2		Construct Shift Register, Universal Shift Register,	K3	2	Lecture With Discussion	BB + ICT
3		Construct Asynchronous counter, Synchronous counter,	K3	2	Lecture With Discussion	BB + ICT
4		Construct Mod-N Counter, binary up/down counter, Ring counter, & Johnson counter.	K3	2	Lecture With Discussion	BB + ICT
5		Explain ROM, PROM, EPROM.	K2	2	Lecture With Discussion	BB + ICT
		TOTAL		9		

Total No. of Classes: 59

Data Structures and Algorithms

Academic Year: 2019-20

Programme: B.Tech

Semester: III

Section: A,B,C& D

Name of the Course: Data Structures and Algorithms

Course Code:V18CST02

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

- CO1:** Explain Sorting and Searching techniques. [K2]
- CO2:** Demonstrate Singly Linked List and Double Linked List. [K3]
- CO3:** Interpret the basic operations on Stacks and Queues.[K3]
- CO4:** Demonstrate Binary Tree and Binary Search Tree.[K3]
- CO5:** Compare Binary trees and self-balanced tree with appropriate examples.[K4]
- CO6:** Illustrate various graph algorithms.[K3]

Text Books:

1. Data Structures, algorithms and applications in C++, SartajSahni, Universities press, Second Edition.
2. Fundamentals of Data Structures in C++, Ellis Horowitz, SartajSahni and Dinesh Mehta, 2nd Edition, Universities Press (India) Pvt. Ltd.

Reference Books:

1. An Introduction to Data Structures with Application, Jean-Paul Tremblay, Paul Sorenson, Second Edition.
2. Fundamentals of Data Structures and algorithms by C V Sastry, Rakesh Nayak, Ch. Raja Ramesh, IK Publications, new Delhi.
3. Data structures using C and C++, Langsam, Augenstein and Tanenbaum, PHI.
4. Problem solving with C++, The OOP, Fourth edition, W.Savitch, Pearson education.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	60	65
CO2	60	65
CO3	60	65
CO4	60	60
CO5	60	60
CO6	60	65

CO1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Discuss CO's and syllabus of the course	K2	1	Lecture	BB
2		Describe different algorithm notations i. Performance analysis: Space Complexity, time complexity ii. Asymptotic notation: Big O, Omega and Theta.	K1	3	Lecture	BB
3		Explain different sorting techniques (Bubble Sort, Insertion Sort, Selection Sort, Quick Sort, Merge Sort, Radix Sort)	K2	4	Lecture with Discussion	BB
4		Describe different searching algorithms (Linear search, Binary search)	K2	1	Lecture with Discussion	BB
5		Discuss about hashing	K2	1	Lecture with Discussion	BB

10**CO2**

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Classify different types of data Structures (Linear data structures and non-linear data structures)	K2	1	Lecture with Discussion	BB+ICT
2		Illustrate polynomial representation using arrays	K3	1	Lecture with Discussion and in class Assignment	BB
3		Compute addition of two polynomials using arrays	K3	1	Lecture with Discussion	BB+ICT
4		Illustrate sparse representation and transpose of sparse matrix using arrays	K2	3	Lecture with Discussion	BB
5		Demonstrate single linked list by node representation and compute different operations on single linked list	K3	4	Lecture with Discussion and in class Assignment	BB
6		Develop different operations like insertion, deletion and display on double linked list.		4	Lecture with Discussion	BB

			K3		and in class Assignment	
7		Demonstrate Circular Linked List	K3	1	Lecture with Discussion	BB+IC T

15

CO3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Describe stack and array representation of stack	K2	2	Lecture with Discussion	BB
2		Describe linked list its representation	K2	1	Lecture with Discussion	BB
3		Illustrate Towers of hanoi with an example	K3	1	Lecture with Discussion and in class Assignment	BB
4		Compute infix to postfix conversion	K3	1	Lecture with Discussion and in class Assignment	BB+IC T
5		Compute expression evolution	K3	2	Lecture with Discussion and in class Assignment	BB+IC T
6		Explain Multistack.	K2	1	Lecture with Discussion	BB
7		Describe Queue and its representation	K2	1	Lecture with Discussion and in class Assignment	BB+IC T
8		Explain queue with linked list representation	K2	2	Lecture with Discussion	BB
9		Develop different operations on queue	K3	2	Lecture with Discussion	BB
10		Explain different applications of Queues	K2	1	Lecture with Discussion	BB
11		Describe Circular queue	K2	1	Lecture with Discussion	BB

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CO4

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Describe Tree and its representation	K1	1	Lecture	BB+CRE
2		Explain Binary Tree, properties of Binary Tree and its representation	K2	3	Lecture with Discussion	BB
3		Develop different operations on Binary tree	K3	2	Lecture with Discussion and in class Assignment	BB
4		Describe different Tree Traversals (recursive and non-recursive)	K1	2	Lecture with Discussion	BB
5		Describe Binary Search Tree and its operations	K3	2	Lecture with Discussion and in class Assignment	BB+ICT

10**CO5**

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Describe different Self-Balanced Trees	K1	1	Lecture with Discussion	BB
2		Explain representation of AVL Tree	K2	2	Lecture	BB
3		Find the height of AVL tree	K3	1	Lecture with Discussion	BB
4		Develop operations like searching, insertion and deletion of an element in AVL Tree	K3	4	Lecture with Discussion and in class Assignment	BB+ICT
5		Define Priority Queue	K1	1	Lecture	BB
6		Explain about Heaps, different types of heaps, properties and operations on heaps	K2	3	Lecture with Discussion	BB

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CO6

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO6	Describe graph, properties of graph, properties of graph, representation of graph and elementary operations on graph	K1	3	Lecture	BB
2		Illustrate the graph traversal using Depth First Search, Breadth First Search techniques.	K3	3	Lecture	BB
3		Explain about Spanning Trees	K2	1	Lecture	BB
4		Construct Kruskal's Algorithm and Prims Algorithm for minimum spanning tree	K3	3	Lecture	BB
5		Construct single source shortest Paths	K3	1	Lecture with Discussion and in class Assignment	BB
6		Construct All pair shortest path algorithm to find shortest graph distances.	K3	1	Lecture with Discussion	BB

12**Total No. of Classes: 74**

Discrete Mathematics

Academic Year: 2019-20

Semester: III

Name of the Course: Discrete Mathematics

Programme: B.Tech

Section: A,B,C& D

Course Code: V18CST03

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, students will be able to:

CO	Course Outcome	Knowledge Level
V18CST03.1	Demonstrate the concepts associated with propositions and mathematical logic.	K3
V18CST03.2	Demonstrate the basic concepts associated with relations functions and their applications.	K3
V18CST03.3	Illustrate algebraic structures and their applications in computer science.	K3
V18CST03.4	Apply techniques of graphs for real-time problems	K3
V18CST03.5	Demonstrate the basic concepts associated with Number Theory	K3
V18CST03.6	Solve recurrence relations using various methods and problems based on combinatorics.	K3

TEXT BOOKS:

1. Discrete Mathematical Structures with Applications to Computer Science, J. P. Tremblay and P. Manohar, 1st Edition, Tata McGraw Hill.
2. Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K. H. Rosen, 7th Edition, Tata McGraw Hill..
3. Discrete Mathematics for Computer Scientists and Mathematicians, J. L. Mott, A. Kandel, T.P. Baker, 2nd Edition, Prentice Hall of India.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	60	60
2	60	60
3	60	60
4	60	60
5	60	60
6	60	60

UNIT-1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Dissemination of Vision, Mission, PEOs,POs,PSOs		1	Lecture	BB
2		Mathematical Logic: Define Statements and their Notations, Connectives	K1	1	Lecture	BB
3		Describe Well Formed Formulas, Truth Tables, Tautologies	K2	1	Lecture with Discussion	ICT
4		Explain equivalence of Formulas	K2	2	Lecture	BB
5		State duality Law, Tautological implications	K1	1	Lecture with Discussion	ICT
6		Convert to normal forms	K2	2	Lecture	BB
7		Explain theory of inference for statement calculus, consistency of premises,	K2	2	Lecture	BB
8		Practice indirect method of proof	K3	2	Lecture	BB
9		Identify statement functions, variables and quantifiers, free and bound Variables	K2	1	Lecture	BB
10		Recognize theory for predicate calculus- Predicates, quantifiers, universe of discourse	K2	2	Lecture	BB
11		Total		15		

UNIT-2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Set Theory and Relations: Define basic concepts	K1	1	Lecture	BB
2		Illustrate operations on binary sets	K2	1	Lecture	BB
3		Use principle of inclusion and exclusion	K3	1	Lecture	BB
4		Describe Relation and properties of binary relations on a set	K2	1	Lecture	BB
5		Sketchout relation matrix and digraph	K3	1	Discussion	ICT
6		Practice equivalence, and partial ordering relations	K3	2	Discussion	ICT+BB
7		Construct hasse diagrams, lattice and state its properties.	K3	2	Lecture	BB

8		Illustrate Functions, bijective functions, composition of functions	K3	1	Lecture	BB
9		Total		10		

UNIT-3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Algebraic Structures: Define algebraic systems and properties of binary operations	K1	2	Lecture	BB
2		Demonstrate semi groups and monoids, homomorphism of semi groups and monoids	K3	2	Lecture	BB
3		Illustrate group, subgroup, abelian group	K2	2	Lecture	BB
4		Total		6		

UNIT-4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Describe basic concepts of graphs, Sub graphs	K1	2	Lecture with Discussion	BB+ICT
2		Illustrate matrix representation of graphs, Adjacency matrices, Incidence matrices	K2	1	Lecture	BB
3		Predict subgraph, isomorphic graphs, paths and circuits	K3	2	Lecture	BB
4		Demonstrate Eulerian and Hamiltonian Graphs, Multigraphs	K3	2	Lecture with Discussion	BB+ICT
5		Use Euler's Formula for Planar Graphs and explain Graph Colouring and Chromatic Number	K3	2	Lecture with Discussion	BB+ICT
6		Total		9		

UNIT-5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Number Theory: Describe properties of integers,	K1	1	Lecture	BB

2		Explain Division theorem, LCM	K2	1	Lecture	BB
3		Compute greatest common divisor using Euclidean Algorithm	K3	2	Lecture	BB
4		Examine testing for prime numbers, fundamental theorem of arithmetic	K3	1	Lecture	BB
5		Illustrate Fermat's Theorem and Euler's Theorem	K3	2	Lecture	BB
6		Total		7		

UNIT-6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO6	Combinatorics Define Basics of Counting	K1	1	Lecture	BB
2		Calculate Permutations, Permutations with Repetitions, Circular Permutations.	K3	1	Lecture	BB
3		Calculate Combinations, Restricted Combinations	K3	2	Lecture	BB
4		Illustrate principle of Inclusion–Exclusion	K2	1	Lecture	BB
5		Recurrence Relations Explain generating functions, Function of Sequences	K2	2	Lecture	BB
6		Solve homogeneous Recurrence relations by substitution and generating functions	K3	2	Lecture	BB
7		Solve Recurrence relations by method of characteristic roots	K3	2	Lecture	BB
8		Solve inhomogeneous recurrence relations	K3	2	Lecture	BB
9		Total		13		

Total No. of Classes: 60

OBJECT ORIENTED PROGRAMMING FOR PROBLEM SOLVING

Academic Year: 2019-20

Semester: III

Name of the Course: OBJECT ORIENTED PROGRAMMING FOR PROBLEM SOLVING

Programme: B.Tech

Section: A,B,C& D

Course Code: V18CST04

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1: Differentiate Procedural Oriented Programming and Object Oriented Programming. [K2]

CO2: Develop object oriented programs on classes and objects. [K3]

CO3: Demonstrate various object-oriented concepts like Constructors,
Destructors & Operator-Overloading. [K3]

CO4: Apply various Object Oriented features like Inheritance and Polymorphism to
solve various computing Problems. [K3]

CO5: Develop programs to handle Exceptions & Files. [K3]

CO6: Describe Generic Programming. [K3]

1. Syllabus

UNIT I: Introduction to Object-Oriented Programming: Introduction to Object-Oriented Programming – Programming Paradigms, Features of Object Oriented Programming, Data Types, Variables, Constants, Operators, Decision Statements & Control Structures, Arrays, Namespace, Default Arguments, Constant Arguments, Inputting Default Arguments, Reference Arguments.

UNIT II: Classes and Objects: Introduction to Classes and Objects : Defining Classes & Objects, Access specifiers, Scope Resolution Operator, Static Member variables, Static Member Functions, Array of Objects. Inline Functions, Overloading Member Functions, Objects as Function Arguments, Friend Functions, Friend Class, Local Class, Empty Class, Nested Classes, Return by Reference.

UNIT III: Object Initialization, Cleanup and Operator Overloading: Introduction to Constructors, Characteristics, Constructor with Default Arguments, Parameterized Constructors, Overloading Constructors, Copy Constructor, Dynamic Constructors and Destructors, Anonymous Objects. Introduction to operator Overloading, Rules for Overloading Operators, Overloading Unary & Binary Operators, this keyword, Constraint on Increment and Decrement Operators, Overloading with Friend Functions, Type Conversions.

UNIT IV: Inheritance and Polymorphism: Base class and Derived class, Single Inheritance, Multiple Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Hybrid Inheritance, Virtual Base Classes, and Constructor in Derived Classes. qualifier classes and inheritance, Significance of Virtual Functions, Early Vs Late Binding, Pure Virtual Functions, Virtual Destructor.

UNIT V: Exception Handling and File Handling: Principles of Exception Handling, Keywords, Exception Handling Mechanism, Multiple Catch Statements, Catching Multiple Exceptions, Re-throwing Exception. File Opening Modes, File Stream Classes, I/O manipulators, Classes for File Handling, Sequential Access Files, Random Access Files, Error Handling Functions.

UNIT-VI: Generic Programming with Templates: Need for Templates, Class Templates, Function Templates, overloading Template Functions. Introduction to Standard Template Library, Sequential Containers & Associative Containers.

3. Text Books

1. Programming in C++, Ashok N Kamthane, 2nd Edition, Pearson.
2. C++ How to Program, Paul J. Deitel, Harvey Deitel, 6th edition, PHI publication.

4. References Books

1. Object Oriented Programming C++, Joyce Farrell, Cengage.
2. Mastering C++, Venugopal, Raj Kumar, Ravi Kumar, TMH.
3. The Complete Reference C++, Herbert Schildt, 4th Edition, McGraw Hill.
4. Object Oriented Programming With C++, R. Subburaj, Vikas Publishing House.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
V18CST04.1	60	60
V18CST04.2	60	60
V18CST04.3	60	60
V18CST04.4	60	60
V18CST04.5	60	60
V18CST04.6	60	60

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Define Object-Oriented Programming	K1	1	Lecture	BB
2		Explain about Features of Object Oriented Programming	K2	1	Lecture+ Discussion	BB
3		Describe about Data Types, Variables, Constants, Operators.	K2	1	Lecture	BB
4		Explain decision Statements & Control Structures with examples	K2	1	Lecture	BB
5		Discuss about Arrays, Namespace, Default Arguments, Constant Arguments	K2	1	Lecture	BB
6		Discuss about Inputting Default Arguments, Reference Arguments.	K2	1	Lecture	BB

7		Demonstrate the example programs using arrays.	K2	2	Lecture	BB
		TOTAL Classes		8		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Define class and object with examples	K1	2	Lecture	BB
2		Explain about Access specifiers, Scope Resolution Operator, Static Member variables	K2	1	Lecture	BB
3		Describe about Static Member Functions, Array of Objects with examples	K3	2	Lecture+ Discussion	BB
4		Illustrate Inline Functions, Overloading Member Functions with example programs.	K3	1	Lecture	BB
5		Explain about Objects as Function Arguments with examples.	K2	1	Lecture	BB
6		Explain about Friend Functions, Friend Class with an example program.	K2	1	Lecture	BB
7		Construct Local Class, Empty Class with an example program.	K3	2	Lecture+ Discussion	BB
8		Construct Nested Classes with an example program and explain.	K3	1	Lecture	BB
9		Prepare a program using the concept Return by Reference	K3	1	Lecture	PPT
		TOTAL		12		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Define Constructor	K1	1	Lecture	BB
2		Explain the characteristics of a constructor.	K2	1	Lecture	BB
3		Demonstrate about Constructor with default arguments, Parameterized constructors,	K3	2	Lecture	BB
4		Illustrate about Overloading constructors with an example.	K3	1	Lecture	BB
5		Construct Dynamic Constructors and Destructors	K3	1	Lecture	BB
6		Explain about Anonymous Objects.	K2	1	Lecture	ICT

7		Explain Operator overloading and its rules with an example.	K2	1	Lecture	BB
8		Demonstrate unary and binary operators	K3	2	Lecture+ Discussion	BB
9		Demonstrate about this keyword, Constraint on Increment and Decrement Operators	K3	1	Lecture	BB
10		Illustrate Overloading with Friend Functions with an example	K3	1	Lecture	BB
11		Explain about Type Conversions	K2	1	Lecture	BB
		TOTAL		13		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Define inheritance	K1	1	Lecture	BB
2		Explain about types of inheritances(Single Inheritance, Multiple Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Hybrid Inheritance)	K2	1	Lecture	ICT
3		Illustrate example programs on Single Inheritance, Multiple Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Hybrid Inheritance	K3	3	Lecture	ICT
4		Demonstrate Virtual Base Classes with example programs.	K3	2	Lecture	BB
5		Demonstrate about Constructor in Derived Classes. qualifier classes and inheritance.	K2	2	Lecture	BB
6		Explain about Early Vs Late Binding, Pure Virtual Functions, Virtual Destructor		1	Lecture	BB
		TOTAL		10		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Define Exception handling	K1	1	Lecture	BB
2		Explain about the keywords in exception handling with an example	K2	1	Lecture	ICT
3		Demonstrate Multiple Catch Statements.	K3	1	Lecture	ICT

4		Illustrate about Catching Multiple Exceptions with an example.	K3	1	Lecture	BB
5		Explain about Re-throwing Exception	K2	1	Lecture	BB
6		Define file and file mode parameters.	K1	1	Lecture	BB
7		Explain about File Opening Modes, File Stream Classes, I/O manipulators	K2	1	Lecture	BB
8		Describe about Classes for File Handling	K2	1	Lecture	BB
9		Explain Sequential Access Files, Random Access Files with suitable examples.	K2	1	Lecture	BB
10		Explain about Error Handling Functions	K2	1	Lecture	BB
		TOTAL		10		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Define Template, Class Template and Function Template.	K1	2	Lecture	BB
2		Demonstrate about Class Templates and function Templates.	K3	1	Lecture	ICT
3		Explain about Standard Template Library.	K2	2	Lecture	ICT
4		Explain about Sequential Containers & Associative Containers.	K2	2	Lecture	ICT
		TOTAL		7		

Total No. of Classes: 60

Digital Electronics Lab

Academic Year: 2019-20

Year/ Semester: III

Name of the Course: Digital Electronics Lab

Programme: B.Tech

Section: A,B,C& D

Course Code: V18ECL04

LESSON PLAN

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO	COURSE OUTCOME	KNOWLEDGE LEVEL
CO1	Apply the Boolean algebra to design digital logic circuits.	K3
CO2	Experiment the behaviour of different combinational logic circuits.	K4
CO3	Experiment the behaviour of different sequential logic circuits	K4
CO4	Construct and troubleshoot simple combinational and sequential circuits	K3

Data Structures and Algorithms Lab

Academic Year: 2019-20

Programme: B.Tech

Semester: III Sem

Section: A,B,C& D

Name of the Course: :Data Structures and Algorithms Lab Course Code: V18CSL02

LESSON PLAN

Course Outcomes (Along with Knowledge Level):

After completion of this course, students will be able to:

V18CSL02.1	Construct programs on sorting and searching techniques.	[K3]
V18CSL02.2	Illustrate various operations on Singly Linked List.	[K3]
V18CSL02.3	Construct programs on Double Linked List.	[K3]
V18CSL02.4	Develop programs on Stacks,Queues and their applications.	[K3]
V18CSL02.5	Implement various operations on Binary Search Tree.	[K3]
V18CSL02.6	Implement various shortest path algorithms.	[K3]

OBJECT ORIENTED PROGRAMMING FOR PROBLEM SOLVING-Lab

Academic Year: 2019-20

Programme: B.Tech

Year/ Semester: III

Section: A,B,C& D

Name of the Course: OBJECT ORIENTED PROGRAMMING FOR PROBLEM SOLVING-Lab

Course Code: R1631049/C309

LESSON PLAN

CO1: Develop Programs on Classes and Objects.	[K3]
CO2: Demonstrate Constructors and Destructors.	[K3]
CO3: Demonstrate Operator-Overloading.	[K3]
CO4: Implement Inheritance and Polymorphism.	[K3]
CO5: Develop programs to handle Exceptions & Files.	[K3]
CO6: Illustrate Generic Programming.	[K3]